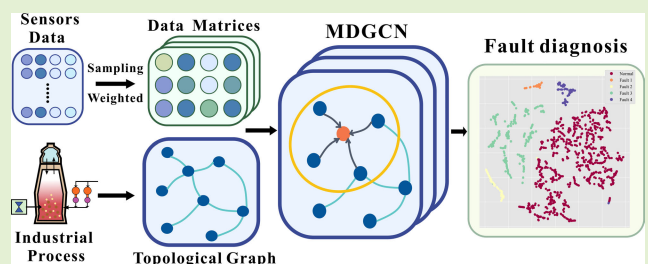


Multichannel Dynamic Graph Convolutional Network-Based Fault Diagnosis and Its Application in Blast Furnace Ironmaking Process

Ping Wu^{1b}, Yixuan Wang^{1b}, Jinfeng Gao, Xujie Zhang^{1b}, Siwei Lou^{1b},
and Chunjie Yang^{1b}, *Senior Member, IEEE*

Abstract—To ensure the operation safety of complex industrial processes, faults that occur in the process should be detected and identified in time to avoid further catastrophic events. Therefore, fault diagnosis plays an indispensable role in the process industry. With the expansion of the production scale of industrial processes, the structural attributed graph is suitable for describing the process data structure due to the complicated interactions between sensor measurements. Graph convolutional networks (GCNs) can identify and capture the relationships between industrial process data in non-Euclidean space by taking graph data with topological structure as input. In this article, a novel fault diagnosis method based on a multichannel dynamic GCN (MDGCN) is proposed. Different from traditional GCNs, the proposed MDGCN assigns different weights to the nodes of the graph. Thus, more useful information about process dynamics is extracted. Particularly, the strategy of multiple isomorphic graph channels is developed to learn feature representations of process data from different levels for fault diagnosis. The capability and efficiency of the proposed MDGCN-based fault diagnosis method are demonstrated through an industrial benchmark of the Tennessee Eastman process (TEP) and a real blast furnace iron-making process (BFIP).

Index Terms—Blast furnace ironmaking process, fault diagnosis, graph convolutional networks (GCNs), multiple isomorphic graph channels.



I. INTRODUCTION

THE modern process industry has become more complex and integrated. With the stringent requirements for product quality and operation safety, fault diagnosis plays a pivotal role in complex industrial processes [1].

Roughly, fault diagnosis methods can be divided into three classes: analytic model-based, expert knowledge-based, and

data-driven based methods [2]. Analytic model-based methods can provide an insightful and sophisticated understanding of the processes. However, the success of model-based methods crucially depends on the establishment of accurate mathematical models [3]. It is very difficult and cumbersome to build accurate mathematical models for complex industrial processes. Expert knowledge-based methods rely heavily on empirical knowledge, which requires long-term available and accurate fault information. On the other hand, data-driven methods construct fault diagnosis models directly from collected historical process data [4]. With the advance of sensor, storage, and computing technologies, data-driven fault diagnosis methods have gathered tremendous attention from academia and industry in recent years [5].

In past decades, traditional statistical learning and machine-learning methods such as Fisher discriminant analysis (FDA), support vector machines (SVMs), the k-nearest neighbor (KNN), and random forest (RF) have been widely used in fault diagnosis methods. However, these methods only extract shallow features from process data. The traditional statistical learning and machine-learning methods are gradually not competent for modeling increasingly complex processes [6]. On the contrary, deep-learning methods can learn deep

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Ping Wu, Yixuan Wang, and Jinfeng Gao are with the School of Information Science and Engineering, Zhejiang Sci-Tech University, Hangzhou 310018, China (e-mail: pingwu@zstu.edu.cn; 202120604131@mails.zstu.edu.cn; jfgao@zstu.edu.cn).

Xujie Zhang, Siwei Lou, and Chunjie Yang are with the College of Control Science and Engineering, Zhejiang University, Hangzhou 310027, China (e-mail: xujie_zhang@zju.edu.cn; swlou@zju.edu.cn; cjiang999@zju.edu.cn).

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features to represent the comprehensive characteristics of process data from deep network structures for addressing complicated tasks [7].

Recently, deep-learning algorithms such as deep belief networks (DBNs), autoencoders (AEs), recurrent neural networks (RNNs), and convolutional neural networks (CNNs) have drawn a lot of attention in fault diagnosis. A DBN is a sophisticated generative model that employs a deep architecture usually by stacking restricted Boltzmann machines. Wang et al. [8] used the restricted Boltzmann machine to fully exploit useful information in the process data and proposed an extended DBN for feature extraction and fault diagnosis. Yu and Yan [9] analyzed the unstable neurons in hidden layers caused by the occurred faults in DBNs for process monitoring. An AE is a typical unsupervised neural network that provides good dimensionality reduction performance by compressing the original input to a low-dimensional feature space. Liu et al. [10] used stacked sparse-denoising AEs with a Softmax classifier to develop an end-to-end fault diagnosis scheme. Jang et al. [11] combined variational AEs and generative adversarial networks to generate features that follow the prior distribution to improve the stability and reliability of fault detection. An RNN is a neural network that contains a feedback mechanism to form a closed-loop structure in the hidden layers. As a classical RNN, long short-term memory (LSTM) networks can avoid vanishing or exploding gradients by introducing “gates.” Han et al. [12] proposed an LSTM-based for fault detection of shipboard vessel components. Peng et al. [13] combined the LSTM model and the causal inference method for solid oxide fuel cell fault diagnosis. Han et al. [14] proposed a fault detection model by determining the optimal number of hidden layer nodes using the iterative method. Inspired by the natural visual perception mechanism, the CNN uses convolutional layers to learn spatial hierarchies of features adaptively. Zhi et al. [15] proposed a CNN-based model that can mine hidden features from processed sensor data to correctly identify faults. Wu and Zhao [16] developed a chemical process fault diagnosis method based on a classical deep CNN (DCNN) model which is composed of convolutional layers, pooling layers, dropout, and fully connected layers. Jiang et al. [17] incorporated a multiscale learning strategy into the traditional CNN architecture to learn high-level fault features at different scales. Xie et al. [18] transformed multiple signal data into three-channel red–green–blue images and then proposed an improved CNN with residual networks for fault diagnosis. However, the CNN only addresses the Euclidean-adjacent feature extraction in the spatial domain. Valuable feature information would be lost by the CNN aggregation approach [19].

For industrial processes such as chemical processes and iron-making processes, the structural attributed graph is suitable for describing the process data structure, since the complicated interactions between sensor measurements exist [20]. Through graph theory, graph neural networks (GNNs) can process data in the graph domain [21]. Among the GNN models, graph convolutional network (GCN) models can make use of the graph structure and aggregate node information from the neighborhoods in a convolutional fashion [22]. Thus, GCNs offer high expressive power for learning

graph representations in a wide range of tasks and applications [23]. Zhao et al. [24] developed a graph convolution DBN for semisupervised fault diagnosis. Chen et al. [25] combined CNNs, GCNs, and attention mechanisms to handle the diversity of working conditions and the lack of sufficient fault samples in fault diagnosis. Sun et al. [26] developed multiscale graph convolution neural networks for machine fault diagnosis. Feng et al. [27] developed a digital twin-enabled domain adversarial graph network for bearing fault diagnosis, where the digital twin model was employed for conducting the dynamic simulation of the bearings. Xu et al. [28] developed a graph-guided CNN model and graph reasoning fusion module to explore the inherent correlations between multisource signals for fault diagnosis of electromechanical systems. Yang et al. [29] combined a GCN model with a gate recurrent unit for the remaining useful life prediction of rolling bearings. Chen et al. [30] used GCN and gated CNN as the associative module and temporal module to integrate the association information and the temporal domain information for providing accurate cooling load prediction. Wu and Zhao [31] integrated the process topology into the GCN model through a transformed graph for fault diagnosis.

Although GCN methods have been successfully applied in fault diagnosis, there is still room for further improvements. First, the conventional GCN models aggregate the information in the temporal domain where the feature weights are the same at different time nodes. However, the dynamic characteristic is inherent in industrial processes [32]. The process data is autocorrelated and cross-correlated. Thus, it would be unreasonable to assign the same feature weights at different time nodes. Second, the GCN can exert a low-pass filtering effect. Yet, improperly increasing the layers of the GCN can exacerbate the smoothing effect of the network [33]. To address these issues, dynamic weights assignment and multiple isomorphic graph channels are developed to enhance the capability of representation learning for fault diagnosis. To deal with these issues, a novel fault diagnosis method based on the multichannel dynamic GCN (MDGCN) is proposed in this work. In the proposed MDGCN, the graph is first constructed using the process topology and knowledge. Next, the Euclidean distance between different sampled process data is calculated and normalized by softmax. These distances are presented as weights to extract the dynamics characteristics of process data. More importantly, the strategy of multiple isomorphic graph channels is employed to construct a multichannel GCN model to learn features from different levels. Finally, a softmax classifier is employed for fault diagnosis.

The main contributions of this work are briefly summarized as follows.

- 1) Time nodes are assigned different weights to aggregate the dynamic information in process data. Through the assignment of different weights at different time nodes, the process dynamics characteristic is further extracted.
- 2) Multiple isomorphic graph channels are integrated into the GNN. Different level features are learned from the constructed graph. By employing multiple isomorphic graph channels, more useful information on process data is explored.

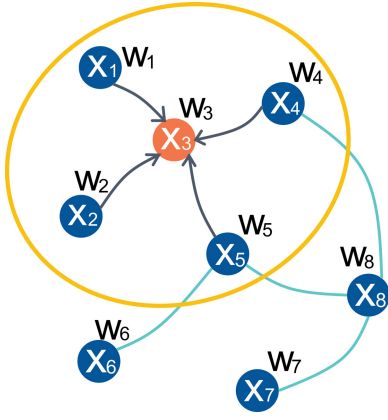


Fig. 1. Spatial-based graph convolutional operation.

- 3) A new fault diagnosis scheme is developed by using the proposed MDGCN and softmax classifier. The superior performance of the proposed MDGCN-based fault diagnosis method is demonstrated by case studies on the industrial benchmark of the Tennessee Eastman process (TEP) and a real blast furnace iron-making process (BFIP), compared to other related methods.

The rest of this article is structured as follows. Section II offers a brief review of the GCN. Section III describes the proposed MDGCN method in detail. In Section IV, the industrial benchmark of TEP and a real BFIP are employed to verify the proposed MDGCN-based fault diagnosis method in comparison with other related methods. Finally, the conclusions are drawn in Section V.

II. REVIEW OF THE GCN

In the GCN, the data is represented as graphs to encode the structural information for modeling the relations among entities. Among different graph construction methods, the local outlier method (LOM) is widely employed [34]. In LOM, the distance of all nodes is calculated to determine the degree of correlation between nodes. Then, the nodes with the closest k Euclidean distances are connected by edges. Specifically, kNN is the most commonly used LOM method in graph construction. kNN is a purely data-driven approach. However, it lacks interpretability. To address this problem, Wu and Zhao [31] integrated process knowledge or topology in graph construction.

GCNs can be considered an efficient variant of CNNs. The virtue of GCNs is that they can aggregate the features of the adjacent nodes in the graph domain. Fig. 1 shows that the GCN can pass information through the edges and aggregate features from neighboring nodes to the central node to update the node state. In GCNs, specific information transfer and aggregation operations are performed by the convolutional layer.

For a given graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, where $\mathcal{V}$ is the set of nodes and $\mathcal{E}$ is the set of edges. $v_i \in \mathcal{V}$ denotes the nodes and $e_{ij} \in \mathcal{E}$ denotes the edges from v_i to v_j . The adjacency matrix $\mathbf{A}$ is an $n \times n$ matrix with $A_{ij} = 1$ if $e_{ij} \in \mathcal{E}$ and $A_{ij} = 0$ if $e_{ij} \notin \mathcal{E}$ [35]. Generally, the information of neighbor nodes around central nodes is required to aggregate. Self-directed

edges are added for constructing graphs. It obtains

$$\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}_n \quad (1)$$

$$\tilde{\mathbf{D}}_{ii} = \sum_{j=1}^n \tilde{A}_{ij}. \quad (2)$$

Here, $\mathbf{I}_n$ is an identity matrix. $\tilde{\mathbf{A}}$ is the adjacency matrix after adding the self-directed edges. $\tilde{\mathbf{D}} = \text{diag}(\tilde{\mathbf{D}}_{ii})$ is the degree matrix with self-directed edges. The spatial-based convolutional operation is formulated

$$\mathbf{X}^{(l+1)} = \sigma \left(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{X}^{(l)} \mathbf{W}^{(l)} \right) \quad (3)$$

where $\mathbf{X}^{(l)}$ is the output of layer l and $\mathbf{X}^{(l+1)}$ is the output of layer $l + 1$. $\mathbf{W}^{(l)}$ is the weight of the layer l and $\sigma(\cdot)$ is the activation function. Through the graph constructed from physical relationships between variables, the process knowledge is included by the adjacency matrix and degree matrix. In addition, the adjacency matrix does not undergo dynamic updates.

III. PROPOSED MDGCN-BASED FAULT DIAGNOSIS SCHEME

A. Dynamic Weights Calculation

Existing GCN models often employ the same weights to capture nodes' structural information, where the weights attached to the data matrix of the current time instant are assumed to be the same as the past period. However, the local structure and dynamic information of process data would be lost. In this work, the weights are dynamically computed through Euclidean-based distance.

The training data $\mathbf{X} = [\mathbf{x}_1 \ \mathbf{x}_2 \ \cdots \ \mathbf{x}_N]^T$ that consists of N observations are collected and normalized. Each observation contains m process variables. Denote the data matrix at time instant t as $\mathbf{X}_t \in \mathbb{R}^{s \times m}$

$$\mathbf{X}_t = [\mathbf{x}_{t-s+1} \ \mathbf{x}_{t-s+2} \ \cdots \ \mathbf{x}_t]^T \quad (4)$$

where s is the number of time lags.

For generality, the calculated Euclidean-based distance is normalized by the Softmax function. Within $\mathbf{X}_t$, the Euclidean distance between $\mathbf{x}_i$ and $\mathbf{x}_j$ is defined

$$d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2. \quad (5)$$

Then, the weight matrix $\mathbf{M} = [m_{ij}] \in \mathbb{R}^{s \times s}$ is defined as

$$m_{ij} = \begin{cases} 1, & \text{if } i = j \\ \frac{\exp(\tilde{d}_{ij})}{\sum_{j=1}^s \exp(\tilde{d}_{ij})}, & i \neq j \end{cases} \quad (6)$$

where $\tilde{d}_{ij} = -(d_{ij}/\bar{d})$. $\bar{d}$ is the mean of d_{ij} .

The relationship between observations $\mathbf{X}_t$ is explored through the defined weight matrix $\mathbf{M}$. Thus, the dynamic characteristic of the process is captured. The matrix $\mathbf{M}$ is used to extract the dynamics of the process data using all the training data at the initialization phase. With the calculated weight matrix $\mathbf{M}$, $\mathbf{X}_t$ is updated as $\mathbf{X}_{t,d}$

$$\mathbf{X}_{t,d} = \mathbf{M}\mathbf{X}_t. \quad (7)$$

It should be noted that there are some abstract nodes such as unit nodes in graph $\mathcal{G}$. These abstract nodes have no actual measured value. By denoting $\mathbf{O} \in \mathbb{R}^{s \times p}$ as a matrix where all elements are set as 0 and p is the number of abstract nodes, the abstract node features are initialized as $\mathbf{O}$. By concatenating the dynamic data matrix $\mathbf{X}_{t,d}$ with the abstract node features $\mathbf{O}$ in horizontal manner, it obtains the feature matrix $\mathbf{X}_{f,t}$

$$\mathbf{X}_{t,f} = [\mathbf{X}_{t,d} \mid \mathbf{O}]^T. \quad (8)$$

Each feature matrix at instant t will be assigned a label $y_t \in \mathbb{Z}$ varying from 0 to c that denotes different process status.

B. Multichannel Dynamic GCN Model

GCNs can act as the function of low-pass filtering. As the number of GCN model layers increases, the smoothing effect of the networks becomes more pronounced [35]. To enhance the ability of representation learning without increasing more layers, the multiple isomorphic graph channels strategy is employed in the work. The architecture of the proposed MDGCN is plotted in Fig. 2. As shown in Fig. 2, information from each node is propagated along topological connections between different units of the complex industrial process. Meanwhile, multiple isomorphic graph channels are employed to enhance the model's ability to learn different representations of the deeper features. Specifically, multiple convolutional kernels are used in this work.

The corresponding degree matrix $\tilde{\mathbf{D}}$ and the adjacency matrix $\tilde{\mathbf{A}}$ are calculated by (1) and (2). It is worth mentioning that $\tilde{\mathbf{A}}$ is symmetric because the constructed graph $\mathcal{G}$ is an undirected graph. For multichannel learning, the model with L graph convolution layers and q filters is initialized, then

$$\mathbf{X}_{t,f,j}^{(l)} = \text{ReLU}(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{X}_{t,f,j}^{(l-1)} \mathbf{W}_{t,j}^{(l)}), \quad j = 1, \dots, q \quad (9)$$

where $l = 1, \dots, L$, $\mathbf{X}_{t,f,j}^{(l-1)}$ is the input feature of the $(l-1)$ th graph convolutional layer, and $\mathbf{X}_{t,f,j}^{(l)}$ is the output feature of the (l) th graph convolutional layer for the j th filter at time instant t . $\mathbf{W}_{t,j}^{(l)} \in \mathbb{R}^{s \times b}$ is the corresponding weight matrix and b is embedded feature dimension. $\text{ReLU}(\cdot)$ represents the Rectified Linear Unit (ReLU) activation function.

Deep representations $\mathbf{h}_j$, $j = 1, \dots, q$ are extracted from l graph convolution layers

$$\mathbf{h}_j = \text{flat}(\mathbf{X}_{t,f,j}^l) = [\mathbf{x}_{t,f,j,1}^l, \mathbf{x}_{t,f,j,2}^l, \dots, \mathbf{x}_{t,f,j,b}^l]. \quad (10)$$

Moreover, the individual flattened features are stacked into $\mathbf{H}$ as

$$\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_q]^T. \quad (11)$$

Then, $\mathbf{H}$ is fed into the multilayer perceptron (MLP) with K layers, and $\mathbf{z}^K$ is the output of the K th layer.

$$\mathbf{H}^k = \text{ReLU}(\mathbf{w}^k \mathbf{H}^{k-1} + \mathbf{b}^k), \quad k = 1, \dots, K \quad (12)$$

$$\mathbf{z}^K = [z_0, z_1, \dots, z_c]^T. \quad (13)$$

A fault diagnosis task can be accomplished as a classification task. To do so, a Softmax layer is employed after the k th full connection layer

$$\mathbf{p} = [p_0, p_1, \dots, p_c]^T = \text{softmax}(\mathbf{z}^K). \quad (14)$$

Algorithm 1 MDGCN Training Procedure

Input: Training dataset $\{\mathbf{X}_t, y_t\}_{t=1}^N$, and the initialized parameters $\mathbf{W}^0, \mathbf{w}^0, \mathbf{b}^0, \mathbf{M}$.
Output: The weight matrix $\mathbf{W}^L, \mathbf{w}^N, \mathbf{b}^N$.

```

1 for epoch = 1, ..., epochmax do
2   for n = 1, ..., N do
3     Calculate the output of convolutional layer  $\mathbf{X}_{f,j}^{(l)}$ 
       by (9);
4     Flatten the output of multi-channels into  $\mathbf{H}$  by
       (11)(12);
5     Feed  $\mathbf{H}$  into the MLP by (13) to obtain the
       output  $\mathbf{z}^k$ ;
6     Convert  $\mathbf{z}^k$  into a probability value vector  $\mathbf{p}$  by
       (14);
7   end
8   Compute the loss function by (15) and optimize the
       parameters with Adam until  $\mathbf{W}^l, \mathbf{w}^k, \mathbf{b}^k$  converge;
9 end

```

The commonly used cross-entropy loss function is adopted

$$\mathcal{L}(\mathbf{W}^l, \mathbf{w}^k, \mathbf{b}^k) = -\frac{y_t \log(p_t) + (1 - y_t) \log(1 - p_t)}{d + 1}. \quad (15)$$

In this work, the model parameters are learned using the Adam optimizer. Moreover, the widely used dropout technique is used to avoid overfitting. The training procedure of the proposed MDGCN model is illustrated in Algorithm 1.

C. Fault Diagnosis Scheme Based on MDGCN

The proposed MDGCN-based fault diagnosis scheme consists of two phases, the offline modeling phase, and the online diagnosis phase. The MDGCN-based fault diagnosis scheme is depicted in Fig. 3.

The proposed MDGCN-based fault diagnosis scheme can be summarized as follows.

1) Offline Training Phase:

- Step1:* Collect and normalize the training data. Construct the graph $\mathcal{G}$ following the process topology and calculate the adjacency matrix $\mathbf{A}$.
- Step2:* Calculate the weight matrix $\mathbf{M}$ to adjust the data matrix $\mathbf{X}_{t,d}$, according to the correlations.
- Step3:* Construct the data matrix $\mathbf{X}_{t,f}$ corresponding to their labels y_t .
- Step4:* Train the MDGCN model using the Adam optimizer and dropout strategy.

2) Online Monitoring Phase:

- Step1:* Collect and normalize real-time samples.
- Step2:* Reconstruct the new data matrix similar to the offline modeling phase.
- Step3:* Update the data matrix using the graph $\mathcal{G}$ and adjacency matrix $\mathbf{A}$ and then feed the data matrix into the trained MDGCN model for prediction.
- Step4:* Determine the faulty type according to the output of the MDGCN.

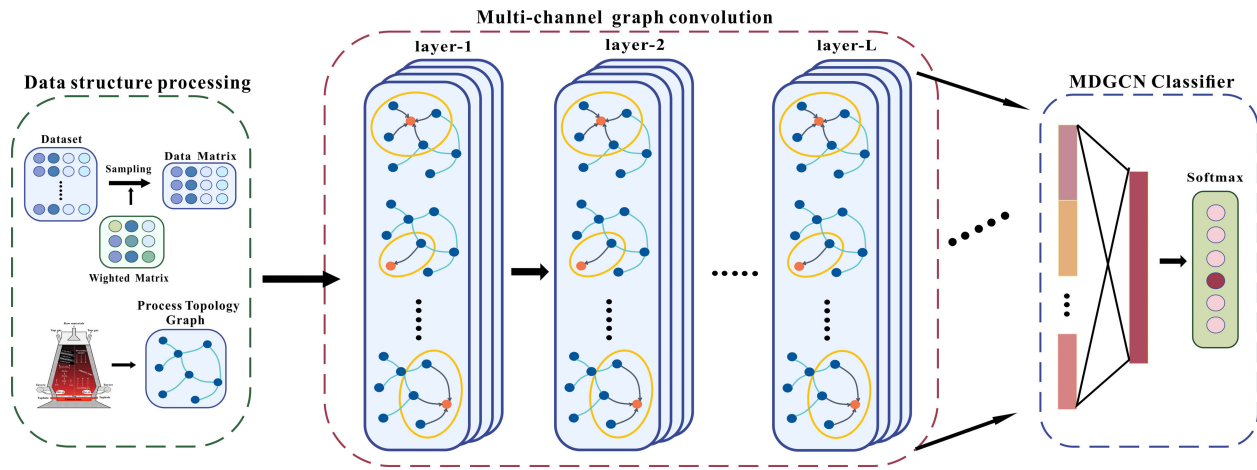


Fig. 2. Architecture of the proposed MDGCN model.

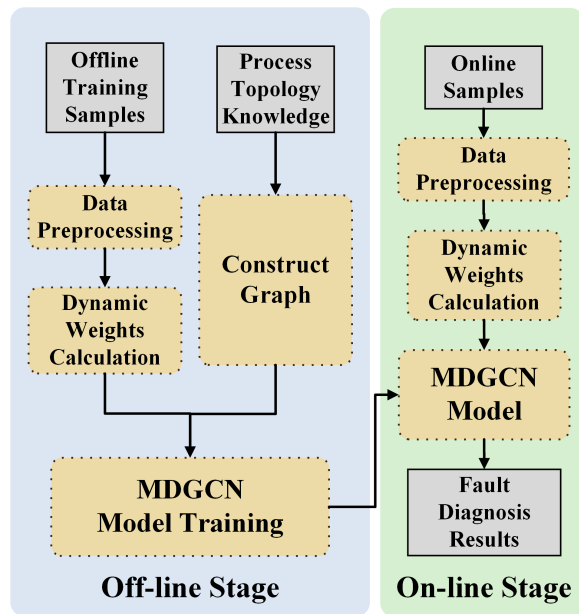


Fig. 3. Framework of fault diagnosis program based on MDGCN.

IV. EXPERIMENTS AND RESULTS

To validate the performance of the proposed MDGCN-based fault diagnosis scheme, experiments on the widely used TEP benchmark and a real blast furnace ironmaking process are carried out. For comparison study, traditional machine-learning methods such as SVM, RF, and deep-learning methods such as DCNN [36], kNN-based GCN (k-GCN) [35], and PTCN [31] are employed. All methods are implemented in a Python 3.8 environment using open-source libraries such as PyTorch and sklearn. For PTCN and MDGCN, the DGL package is also used. The working environment is Windows 10, the GPU is GeForce GTX 1080, and the CPU is Intel¹ Xeon¹ Bronze 3104.

A. TEP Benchmark Case

The TEP is a widely used benchmark that was developed based on an actual chemical process, for evaluating the performance of process monitoring and fault diagnosis. The TEP consists of five operating units, a reactor,

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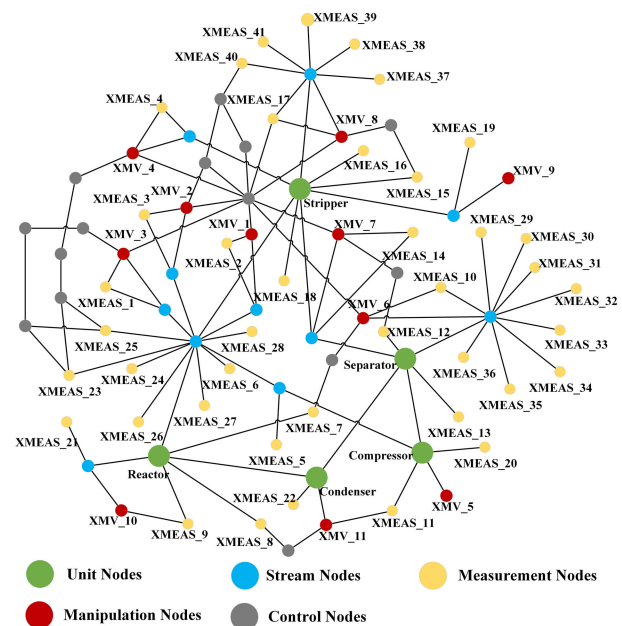


Fig. 4. Established graph of TEP.

a condenser, a vapor-liquid separator, a recycle compressor, and a product stripper. The process contains 12 manipulated variables (XMV1-12) and 41 measured variables (XMEAS1-41) with a sampling period of three minutes. In this work, 52 variables (XMV(12) excluded) are employed. 21 different fault scenarios (Faults 1–21) were simulated. In this work, a total of 18 fault types excluding faults 3, 9, and 15 are selected for comparative study. More details about the TEP can be found in [37]. In this work, the benchmark data found in <http://web.mit.edu/braatzgroup/links.html> are used. The dataset includes 3380 samples and 800 samples for each type of fault. The datasets are divided into 70% of the training set and 30% of the test set. The graph of TEP $\mathcal{G}^{\text{TE}}$ is established using the method introduced in Section II. In $\mathcal{G}^{\text{TE}}$, there are 82 nodes and 111 edges. Fig. 4 displays the diagram of $\mathcal{G}^{\text{TE}}$.

To evaluate the fault diagnosis performance, two indices including fault diagnosis rate (FDR) and accurate classification rate (ACR) are used [31]. The FDR and ACR are defined

TABLE I

ACR RESULTS WITH DIFFERENT LAYERS

Depth	1	2	3	4	5
ACR	0.959	0.971	0.973	0.977	0.961

TABLE II

ACR OF MDGCN WITH DIFFERENT CHANNELS

Channels	1	2	3	4	5
ACR	0.977	0.981	0.985	0.987	0.982

TABLE III

STRUCTURE OF THE MULTICHANNEL GCN ON TEP

Name	Channel	Structure
Convolution	4	Gc(16)-Gc(16)-Gc(16)-Gc(16)
FC Layer	/	FC(82×16,0.5)-FC(300,0.5)-FC(19)

below

$$\text{FDR} = \frac{n_{ct}}{n_c} \times 100\% \quad (16)$$

$$\text{ACR} = \frac{\sum_c n_{ct}}{\sum_c n_c} = \frac{\sum_c n_{ct}}{n_{\text{total}}} \quad (17)$$

where n_c denotes the total number of test samples for class c , and n_{ct} denotes the number of correctly classified class c samples. The ACR represents comprehensive classification accuracy. A higher ACR indicates that the performance is superior.

For MDGCN, the time lag is set as 10. It is critical to select the appropriate model structure for deep learning tasks. To obtain a satisfactory structure of the proposed MDGCN model, the number of layers is determined by evaluating the performance of the MDGCN model which does not employ a multichannel strategy. Table I lists the ACRs with different layers. From the data in Table I, it can be found that the best depth should be selected as 4. The increase in the channel number can improve performance, but more channels do not necessarily lead to better results. The reason could be the more templates we use, the more complex our model would be [38]. Thus, the number of channels should be selected appropriately. The selection of the number of channels is according to experience and experiment. The ACRs of MDGCN models with different channels are listed in Table II. As shown in Table II, the best number of channels is 4.

The architecture of the MDGCN is shown in Table III. Gc(m) represents a convolutional graph layer with a hidden state embedding dimension of m , and FC(m, n) is a fully connected layer with m neurons using a dropout ratio of n . According to the preexperiment, the number of channels of graph convolution is set as 4. Thus, four different deep features are learned simultaneously. All four channels have the same structure. The number of convolution layers for DCNN, PTCN, k-GCN, and MDGCN is set as 4. The number of neighbors for k-GCN is set to 5. These models are trained using the ReLU activation function, cross-entropy loss functions, Adam optimizer, and mini-batches with a size of 128. The learning rate is set as 0.001.

The diagnostic results are listed in Table IV. As shown in Table IV, although FDRs of normal, faults 1–7 and 18 can reach 90%. However, SVM has a low ACR. Compared to SVM, RF can offer better FDR for scenarios. The ACR of RF

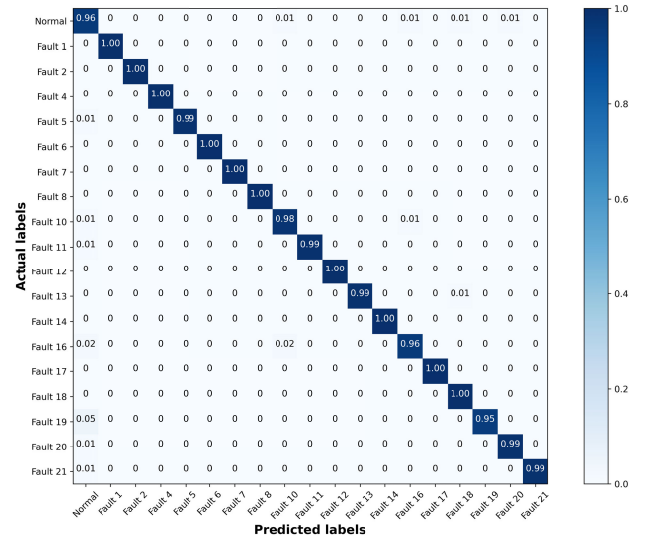


Fig. 5. Confusion matrix of the MDGCN-based fault diagnosis.

is still below 90%. Different from the shallow learning methods, deep-learning methods such as DCNN, k-GCN, PTCN, and MDGCN have dramatically improved performance. Moreover, as shown in Table IV, the ACR of the proposed MDGCN method reaches 98.8% which is the highest among the comparable methods. It is 2.1% higher than DCNN, 3% higher than k-GCN, and 1% better than PTCN. On the other hand, it can be found that the ACRs of PTCN and MDGCN are higher than k-GCN. By introducing process knowledge to construct the graph, both PTCN and MDGCN provide better performance than k-GCN which builds the graph in a data-driven way.

To further analyze the performance of MDGCN, the confusion matrix is plotted in Fig. 5. In accordance with the data in Table IV, MDGCN correctly classifies almost entirely of samples, except for a small number of normal samples.

B. Real BFIP Case

The BFIP is one of the most complex industrial processes. There are five main subsystems such as blast furnace, feeding, hot air, pulverized coal injection, and exhaust gas treatment to generate molten iron. In the BFIP, raw materials and fuels, hot air, and pulverized coal are fed into the blast furnace from the furnace top and tuyere, respectively. A series of complex physical and chemical reactions happen under high temperatures and pressure. Molten iron is generated from the blast furnace along the tap hole. Since the harsh working conditions, the BFIP often suffers from various abnormal furnace conditions such as channeling, collapse, and hanging [39]. If these abnormalities are not detected and diagnosed timely and accurately, it will cause degradation of the quality of molten iron, even threats to the safety of equipment and personnel. Thus, fault diagnosis is crucial to ensure the operation safety of the BFIP.

To build the graph of BFIP for fault diagnosis, BFIP can be divided into four main units including the blast furnace main body, the top of the blast furnace, the pulverized coal injection system, and the hot blast system according to the mechanism of the iron-making process, as shown in Fig. 6.

From the expert knowledge, ten main process measurements are selected for identifying the BFIP working status. Table V describes the selected ten process variables. In this study, data

TABLE IV
COMPARISON RESULTS ON TEP

Fault Type	SVM	RF	DCNN	k-GCN	PTCN	MDGCN
Normal	0.992	0.966	0.946	0.904	0.956	0.969
Fault 1	0.994	0.985	1	0.996	1	1
Fault 2	0.958	0.951	0.996	0.986	0.996	1
Fault 4	0.995	0.973	0.983	0.978	0.991	0.996
Fault 5	0.994	0.714	0.965	0.906	0.996	0.996
Fault 6	1	0.991	0.993	1	1	1
Fault 7	1	0.991	0.994	1	1	1
Fault 8	0.254	0.852	0.987	0.987	0.996	1
Fault 10	0.053	0.843	0.929	0.938	0.965	0.978
Fault 11	0.001	0.708	0.992	0.965	0.988	0.996
Fault 12	0.381	0.840	0.966	0.972	0.993	0.996
Fault 13	0.697	0.913	0.944	0.966	0.987	0.991
Fault 14	0.041	0.970	0.996	0.996	1	1
Fault 16	0.209	0.823	0.948	0.868	0.965	0.959
Fault 17	0.883	0.925	0.992	0.992	0.996	0.996
Fault 18	0.905	0.873	0.967	0.977	0.982	0.996
Fault 19	0.004	0.705	0.983	0.881	0.943	0.954
Fault 20	0.736	0.698	0.970	0.963	0.991	0.991
Fault 21	0.618	0.881	0.986	0.990	0.991	0.995
ACR	0.594	0.871	0.967	0.958	0.978	0.988

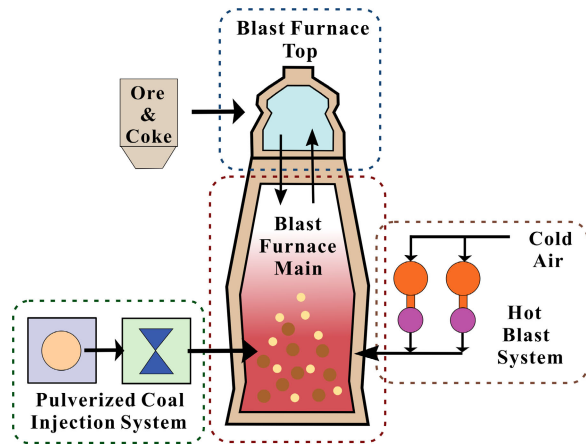


Fig. 6. Divided units of BFIP.

TABLE V
MONITORED VARIABLES FOR THE BFIP FAULT DIAGNOSIS

No.	Variable(Unit)
V1	CO Volume (%)
V2	H ₂ Volume (%)
V3	CO ₂ Volume (%)
V4	Enriching oxygen flow (m ³ /h)
V5	BF top gas pressure (MPa)
V6	Cold blast pressure (MPa)
V7	Total pressure drop (MPa)
V8	Hot blast pressure (MPa)
V9	Drag coefficient
V10	Actual coal injection rate (Ton/h)

were collected from the #2 BF in Liuzhou Steel Company Ltd., which is the largest working BF in the Guangxi Province of China. Specifically, 4667 samples including 125 collapse samples, 216 furnace temperature rise samples, 1057 channeling samples, 245 furnace temperature down samples, and 3018 normal samples were collected. Following the procedure described in Section II, the graph $\mathcal{G}^{\text{BF}}$ is derived as shown in Fig. 7. $\mathcal{G}^{\text{BF}}$ consists of 20 nodes and 23 edges.

According to the experience and validation, the architecture of the MDGCN model is listed in Table VI. For comparison study, SVM, RF, DCNN, and PTCN are employed. For a

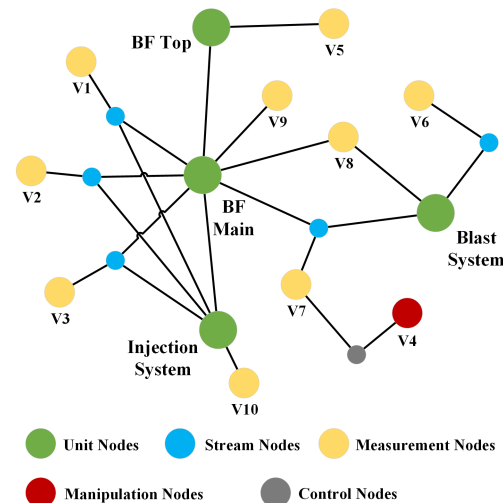


Fig. 7. Diagram of the BFIP graph.

TABLE VI
STRUCTURE OF THE MDGCN MODEL FOR BFIP FAULT DIAGNOSIS

Name	Channel	Structure
Convolution	5	Gc(8)-Gc(8)-Gc(8)-Gc(8)
FC Layer	/	FC(200,0.5)-FC(5)

TABLE VII
TRAINING AND TESTING TIME USED IN THE BFIP

Model	DCNN	k-GCN	PTCN	MDGCN
training time(s)	8.28	17.83	10.45	18.32
testing time(s)	0.002	0.011	0.009	0.047

fair comparison, the number of convolution layers for DCNN, k-GCN, PTCN, and MDGCN are all four layers. The number of neighbors for k-GCN is set to 2. For MDGCN, the time lag is set as 10.

Fig. 8 shows the testing ACR with different training epochs. MDGCN has a better coverage rate and higher ACR than DCNN, k-GCN, and PTCN. And the ACR of MDGCN nearly reaches 100%.

TABLE VIII
COMPARISON RESULTS ON BFIP

Fault Type	SVM	RF	DCNN	k-GCN	PTCN	MDGCN
Normal	0.977	0.991	0.991	0.953	0.999	0.997
Fault 1 (Collapse)	0.135	0.729	0.879	0.958	0.971	0.972
Fault 2 (Furnace temperature rise)	0.846	0.923	0.885	0.976	0.986	1
Fault 3 (Channeling)	0.958	0.968	0.991	0.975	0.994	1
Fault 4 (Furnace temperature down)	0.675	0.891	0.944	0.954	0.944	1
ACR	0.718	0.901	0.938	0.958	0.979	0.994

TABLE IX
RESULTS UNDER NOISE ON BIFP

Model	RF	DCNN	k-GCN	PTCN	MDGCN
ACR	0.826 ± 0.024	0.925 ± 0.033	0.939 ± 0.012	0.967 ± 0.015	0.987 ± 0.003

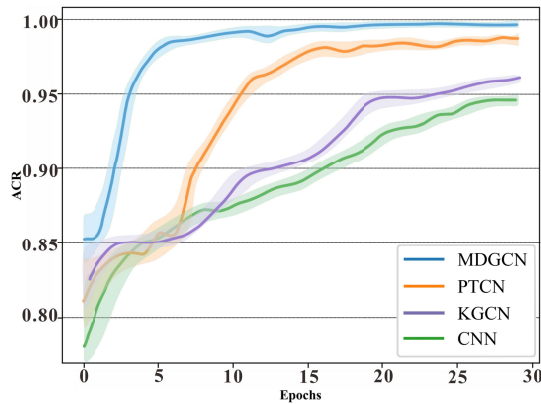


Fig. 8. Testing ACRs with the number of training epochs.

Table VII lists the training and testing time used in the BFIP case. Compared to other methods, the proposed MDGCN model would cost more time in the training phase since it involves the initialization of dynamic weights and the updates of multichannels. From the data in Table VII, it can be found that the training and testing times of the proposed MDGCN are higher than other methods. The training time and testing time of the MDGCN are 18.32 and 0.047 s, respectively. However, the computational cost of the proposed MDGCN is at the same level as other methods such as PTCN and k-GCN. In addition, the proposed MDGCN does not require more samples in the training phase.

To compare the performance of DCNN, k-GCN, PTCN, and MDGCN, the t-distributed stochastic neighbor embedding (t-SNE) technique is utilized to visualize the learned features. The clusters obtained by t-SNE are plotted in Fig. 9. As shown in Fig. 9, it can be observed that the clusters of the features extracted by MDGCN can be better separated than DCNN, k-GCN, and PTCN. Therefore, the classification accuracy of MDGCN is higher.

The diagnostic results are listed in Table VIII. As shown in Table VIII, SVM provides poor fault diagnosis performance, where the ACR is only 71.8%. RF can offer better performance, where the ACR reaches 90.1%. Compared to SVM, RF can efficiently capture the nonlinearity of BF. It can also be observed that deep-learning methods such as DCNN, k-GCN, PTCN, and MDGCN can derive superior performance than shallow learning methods such as SVM and RF. Their ACRs are higher than that of RF. When the data dimension is

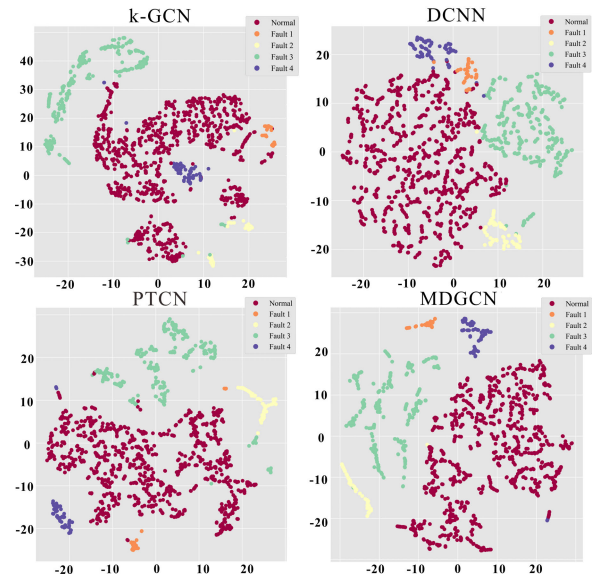


Fig. 9. t-SNE visualization of features learned by different models.

not high, k-GCN without graph topological knowledge works better than CNN.

To further compare the MDGCN model to other models, Gaussian noise with a mean of 0 and a standard deviation of 0.3 is added to the normalized variables (V1–9). Then, the generated dataset is used to test the robustness of different models. Each model was run 30 times using the randomly generated datasets. The mean and standard deviation of the testing results are listed in Table IX. From the data in Table IX, it can be concluded that the proposed MDGCN can provide the highest ACR and lowest standard deviation. Thus, it indicates that the MDGCN model has good robustness.

In the proposed MDGCN model, two important improvements are developed, including assigning dynamic weights and employing the strategy of multiple isomorphic graph channels. To illustrate the benefits brought by these improvements, ablation experiments are carried out. To do so, we compare three models such as PTCN, dynamic GCN, and multichannel GCN with the MDGCN model. PTCN only considers the process topology. Dynamic GCN takes the process topology and dynamic weights into account. Multichannel GCN integrates the process topology and employs the strategy of multiple isomorphic graph channels. The proposed MDGCN model takes advantage of the process topology, dynamic weights,

TABLE X
ABLATION EXPERIMENTAL RESULTS OF MDGCN FAULT DIAGNOSIS METHOD ON BFIP

Fault Type	PTCN	Dynamic GCN	Multi-channel GCN	MDGCN
Normal	0.999	0.998	0.996	0.997
Fault 1 (Collapse)	0.971	0.937	0.972	0.972
Fault 2 (Furnace temperature rise)	0.986	0.986	1	1
Fault 3 (Channeling)	0.994	0.995	1	1
Fault 4 (Furnace temperature down)	0.944	1	0.995	1
ACR	0.979	0.983	0.993	0.994

and multiple isomorphic graph channels simultaneously. The ablation experimental results are given in Table X. From the data in Table X, it can be found that fault 4 can be correctly identified by assigning dynamic weights to construct the graph. The multichannel convolution operation is also helpful to identify faults 2 and 3. Although the classification accuracy of normal samples is slightly lower than PTCN and dynamic GCN, the overall accuracy of MDGCN is higher.

V. CONCLUSION

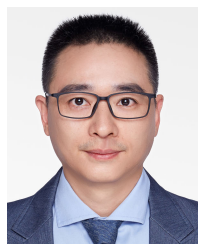
In this article, a novel fault diagnosis method based on MDGCN is proposed. The proposed MDGCN incorporates the process topology knowledge and assigns different weights to construct the graph for extracting more useful information. More importantly, multiple isomorphic graph channels are employed to capture the representation of process data from different levels. The effectiveness of the proposed MDGCN is verified through the industrial benchmark of TEP and a real BFIP. Although the proposed MDGCN-based fault diagnosis method offers superior performance over other related methods, there are still some issues that should be addressed in future work.

- 1) In the proposed MDGCN model, the graph is mainly constructed using process knowledge. To improve the efficiency of graph construction, the data-driven and knowledge-driven methods could be combined in further study.
- 2) To enhance the ability to extract temporal information, more neural network architectures such as LSTM networks and attention mechanisms can be taken into consideration in the framework of MDGCN in future research.
- 3) Since the process topology knowledge is fully embedded in the MDGCN model, the root cause analysis could be more accessible. Future work will focus on the root cause analysis using the proposed MDGCN model.

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Ping Wu received the B.S. and Ph.D. degrees in control theory and control engineering from Zhejiang University, Hangzhou, China, in 2003 and 2009, respectively.

He is currently an Associate Professor with the School of Information Science and Engineering, Zhejiang Sci-Tech University, Hangzhou. His major research interests include fault diagnosis, machine learning, and industrial intelligence.



Yixuan Wang received the B.S. degree in automation from Hangzhou Dianzi University, Hangzhou, China, in 2021. He is currently pursuing the M.S. degree in control science and engineering from the School of Information Science and Engineering, Zhejiang Sci-Tech University, Hangzhou.

His current research interests include multivariate statistical analysis, deep learning, and fault diagnosis.



Jinfeng Gao received the B.S. degree from the Hebei Institute of Science and Technology, Shijiazhuang, China, in 2000, the M.S. degree from the Zhejiang University of Technology, Hangzhou, China, in 2003, and the Ph.D. degree from Zhejiang University, Hangzhou, in 2008.

She is currently a Professor with the School of Information Science and Engineering, Zhejiang Sci-Tech University, Hangzhou. Her research interests include fault detection and diagnosis, networked control, and multiagent systems.



Xujie Zhang received the B.S. degree in mechatronics engineering and the M.S. degree in control science and engineering from Zhejiang Sci-Tech University, Hangzhou, China, in 2019 and 2022, respectively. He is currently pursuing the Ph.D. degree in control science and engineering with the College of Control Science and Engineering with Zhejiang University, Hangzhou.

His current research interests include deep learning, process monitoring, and fault diagnosis.



Siwei Lou received the B.Eng. degree in automation and the M.Sc. degree in control science and engineering from the School of Mechanical Engineering and Automation, Zhejiang Sci-Tech University, Hangzhou, China, in 2018 and 2021, respectively. He is currently pursuing the Ph.D. degree in control science and engineering with the College of Control Science and Engineering, Zhejiang University, Hangzhou.

His current research interests include statistical machine learning, deep learning, process monitoring, and fault diagnosis.



Chunjie Yang (Senior Member, IEEE) received the B.Eng. degree in machine design, the M.Eng. degree in fluid transmission and control, and the Ph.D. degree in industrial automation from Zhejiang University, Hangzhou, China, in 1992, 1995, and 1998, respectively.

He is currently a Qiushi Distinguished Professor with Zhejiang University, where he is also a Professor with the College of Control Science and Engineering. His current research interests include modeling, control, and fault diagnosis

for industrial processes, smart energy, and intelligent manufacturing. He has authored and coauthored more than 100 articles in these areas.